

ONE STEP BEYOND: MAKING B2B DECISIONS WITH BUSINESS ACTIVITY MONITORING AND SOCCER^{*}

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Abstract - We present a formal method to execute Business Activity Monitoring (BAM) on a Small or Medium Sized Business (SMB) that is undergoing re-engineering towards process-based workflow operations. On these enterprises, real-time access to critical performance indicators improves the speed and effectiveness of their overall logistical machinery. Businesspeople will be defined as actors in uncertain and conflictive situations. The dynamics of real-life negotiation poses challenges when creating models of how people interact and settle down strategic differences. We offer a functional analogy between collaborative strategies in a soccer match and B2B processes that feed on BAM data. We characterize negotiations using Q-Learning on actions that feed on BAM knowledge when searching for a common goal. We present Soccer and BAM-B2B demo software. Along the way, we pay special attention to the needs of SMB's, both in Mexico, and abroad, and thus concentrate on designing future products for Business Process Management (BPM).

Keywords: Soccer, Business Activity Monitoring, Agents, Negotiation, Q-Learning.

1 Introduction

In the world of Business-to-Business (B2B), and Mobile Commerce (m-Commerce) [1], a *business process management* (BPM) [2,3,4,5] tool offers comprehensive and automated supply chain process support. In other words, a set of business processes run in their local sites but interact using inter-enterprise conversation schemas, conversations, or choreographies [6]. As a result, a partial-to-total integration and automation of information, logistical, and strategic systems will take place as the workflow makes a progress over the supply chain. According to Gartner Research [7], a set of desirable feature in all BPM tools include:

- ✓ Support of business process flow for *human-to-human related tasks*, rather than just system-to-system and human-to-system.
- ✓ Business activity monitoring (BAM) capabilities.

It is especially interesting that human-to-human tasks are considered in these requirements. The authors hereby expand this notion to *intra-enterprise-human-to-human activity* flow, because, as we have mentioned in [9], the dynamic nature of modern business environments includes a decision-making process that is frequently characterized by conflicting situations. Frictions derived from opposing interests, as well as competition, and other troublesome situations resulting from unlawful practices are all common, and expected, in human relations. Therefore, we name these: *intra-enterprise-human-to-human activity*, and point out that they are being addressed by industrial BPM trends. Such activity locks into place alongside commonplace work among systems, and human-system workflow.

On the other hand, *Business Activity Monitoring* (BAM) is an emerging technology aimed at providing real-time access to critical business performance indicators which impact the overall performance of the company. BAM technologies monitor business activity and trigger alerts as well as simple or complex events based on business rules. Currently, it is posed as a middleware solution in the process of being integrated to enterprise applications. BAM is still in its early stages and promises to deliver practical results over the next five years. It is expected that leading enterprise software vendors will have incorporated BAM into their products by 2006 [8]. BAM is a form of BPM.

Nevertheless, BPM has typically been out of reach for small and midsize businesses (SMB's). It is very likely that BAM will follow the same corporate pattern. However, a huge potential lies in delivering these technologies to SMB's. Over 99% of enterprises in the EU are SMB's. They employ over 70% of the total labor force in this part of the world. Likewise, in Japan, 75% of all manufacturing employment is in SMB's. In Latin America, over 85% of all

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enterprises are SMB's and constitute the core of the economy, the GNP, and of all bases for taxation.

Nowadays, most of the 40 million small and midsize businesses (SMB's) are so underteched, but constitute so much of a potential worldwide, that even Microsoft is planning to invest \$2 billion on them over the next few years [9]. Specifically, SMB's require an entry channel to the use of mainstream technology such as Business Intelligence (BI), Supply Chain Management (SCM) and Customer Relationship Management (CRM). It is also a fact that small companies must [partially] become online in order to survive current Business and IT trends.

Therefore, our research is aimed at providing an ontological foundation for enabling BAM in wireless and m-commerce scenarios. We are especially interested in making a contribution which can later be ported to SMB's in Mexico, Latin America, and the world.

2 Business Processes are Soccer Matches

This paper presents a novel approach to characterizing and visualizing BAM and BPM technology. Firstly, we use a B2B-Soccer metaphor we previously presented in [10]. Hereby, we proved that a set of soccer players collaborating to reach their goal: scoring, face problems and contradictions when deciding the best plays and strategy. Likewise, a set of employees in a firm will have to work as a team to reach their goal: score more contracts, and increase revenues, but will have to decide upon professional and personal differences along the way. The metaphor merged B2B models by Ravi Kalakota [11], and a soccer field (see Figure 1).

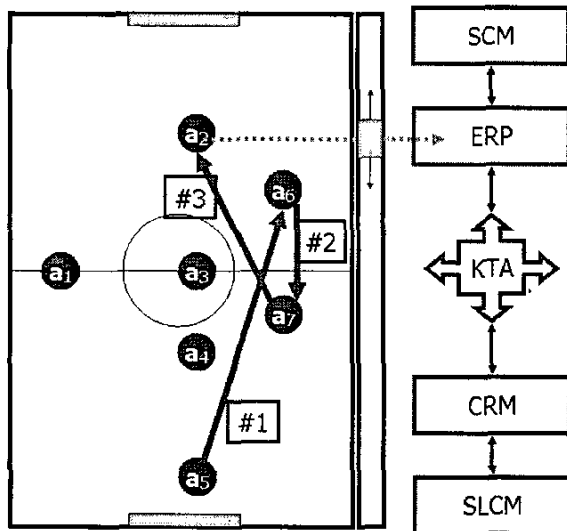


Figure 1, Soccer-B2B Metaphor

In the metaphor, soccer players pass the ball after deciding who the best receptor is given a decision-making calculation based on a risk vs. utility Q-Learning positioning function, as shown on (1).

$$r(BA, a_{mn}) = Utility(BA, a_{mn}) - Risk(BA, a_{mn}) + Positioning(BA, a_{mn}) \quad (1)$$

During such calculations, data must be fed into (1), given specific operational parameters. Consider equation (2). "Sanborn's" is a popular Mexican retailer.

$$r(Sell, a_{\{Sanborn's\}\{k\}}) = \{PS(Sanborn's, k) - CM(a_{\{Sanborn's\}\{k\}})\} - \{EX(Sanborn's) + CS(Sanborn's, Inv_{passive})\} + \{BL(Sanborn's, svc, spp)\} \quad (2)$$

A given sales agent (a_k) works for the retailer. Sanborn's will obtain reward (r) using utility based on profit (PS), commissions by sales agents (CM), general expenses (EX), cost of passives (CS), and indicators on service vs. support (BL). Eventually, these data will serve for the purpose of calculating reward on a daily basis. Reward means how fruitful a given combination of factors (PS, CM, EX, CS, BL) have been after a series of business processes have been executed, that is, after a sales agent (a) has finished a selling operation, profit has been obtained, commissions have been charged, passives have been modified, expenses have been reported, and balances have been issued. Witness equation #3 below; sales agents 2, 3, 4 & 5 have each performed their duties with mixed results.

$$\begin{aligned} r(Sell, a_{\{Sanborn's\}\{2\}}) &= \{PS(20) - CM(1)\} - \{EX(40) + CS(10)\} + \{BL(8)\} = 19 - 50 + 8 = -23 \\ r(Sell, a_{\{Sanborn's\}\{3\}}) &= \{PS(22) - CM(3)\} - \{EX(18) + CS(7)\} + \{BL(8)\} = 19 - 25 + 8 = 2 \\ r(Sell, a_{\{Sanborn's\}\{4\}}) &= \{PS(23) - CM(3)\} - \{EX(15) + CS(6)\} + \{BL(8)\} = 20 - 21 + 8 = 7 \\ r(Sell, a_{\{Sanborn's\}\{5\}}) &= \{PS(25) - CM(5)\} - \{EX(10) + CS(5)\} + \{BL(8)\} = 20 - 15 + 8 = 13 \end{aligned} \quad (3)$$

So that a state action matrix may be obtained (see equation 4).

$$a1: \begin{bmatrix} a2 & 19 & 50 & 8 \\ a3 & 19 & 25 & 8 \\ a4 & 19 & 20 & 8 \\ a5 & 20 & 15 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} -23 \\ 2 \\ 7 \\ 13 \end{bmatrix} \quad (4)$$

The best reward corresponds to the actions by proposal a5 (i.e. the larger quantity). Henceforth, the next agent (number 1) will be able to decide what strategy provides a better reward, in terms of business factors.

Such calculations must be updated frequently so that the model is kept to date. In our simulator software, this was easily obtained from a local database. The simulator is shown in Figure 2.

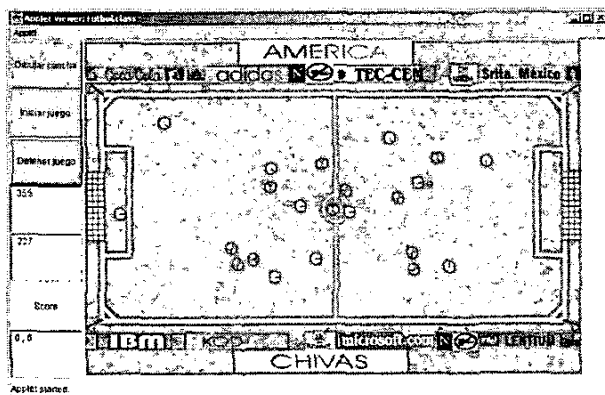


Figure 2, B2B Simulator

However, in a real-life situation, data will not be as accessible. Furthermore, accessibility will also suffer from imprecision and possible loss. Thus, BAM is proposed as a suitable tool for providing feedback in a more accurate and reliable manner.

3 BAM as Feedback on ORMS

The authors then propose that BAM may be implemented using a cost-effective IT solution for providing data on everyday Operational Resource Management Systems (ORMS) [11]. From this point on, we will speak of ORMS as "reward management" in terms of equations (2), (3) and (4) in the previous section. BAM is hereby implemented building over architectures proposed by Cheng and Gartner in [4, 5] and shown in Figure 3.

Events will be generated at a number of remote business stations. Such stations may be accounting PC's, payroll mainframes, inventory management workstations, wireless Points of Sale (POS), etc. They will be transferred over to a BAM monitor central in XML stream formats using standard UDP or TCP socket connections. Events will be of two

types: (1) Technical, and (2) Business. Technical events are those created by machines, computer operators, IT staff and any other employee dealing with databases, telecommunications, printing, or man-machine interaction services. Examples are: "Commit", "Open Socket, Port 7786", "Close File", "Shutdown Server, 18:34hs", etc. Business events will be higher level, for example: "Contact New Customer", "Open Order", "Back Order", "Confirm Payment", etc. Both types of events will be "absorbed" and filtered by a set of business rules written to allow a Process Manager Administrator (PMA) to create views of the chain of events being captured.

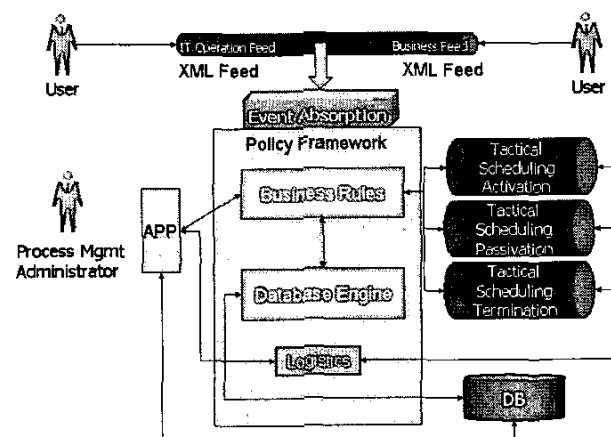


Figure 3, BAM Architecture

For example, the PMA may want to view activity on payments ONLY. At another time, she may want to witness just orders, inventories and payments. Event filtering may thus be activated (an event set is allowed to go through), passivated (an event set is allowed to go through on time intervals $\langle t_0, t_1 \rangle$ or given a set of conditions $\langle c_0, c_1 \dots c_k \rangle$), or terminated (an event set is prevented from going through). A database engine will process all authorized events by adding time stamps, host names, IP addresses, and other information that arrives over the XML stream. The final versions of event registers will be added to a relational database (DB). From this point on, the PMA will be able to report business activity over time from her administrative APP.

4 Q-Learning on BAM Data

BAM monitoring as described in the previous section is defined to be tactical (i.e. short-term). However, the PMA may be responsible for pinpointing new strategic actions to improve the overall business performance. Thus, her APP in Figure 3 may now use the BAM feed stored in the database (DB) for making a comprehensive view of possible rewards given specific actions executed by the company's staff.

Consider equation (2). We will now modify it so a set of operational parameters is obtained from the BAM monitor.

Hereby, a value $[i]$ is calculated as follows:

$$i \in \dim(\text{BAM}[\text{Sanborn's}, a, k]) \quad (4)$$

This value is obtained by executing a query to the database in Figure 3. It selects a number of BAM registers detected from Sanborn's, as a trace of activity by any sales agent k . Next, parts of (2) are seen as similar queries. For example, $[PS(\text{Sanborn's}, i)]$ is a query to extract profits from sales by agent $[i]$. Likewise, $[CM(a_{\text{Sanborn's}}, i)]$ extracts commissions paid to agent $[i]$ given the aforementioned sales, $[EX(\text{Sanborn's})]$ looks for all company expenses, $[CS(\text{Sanborn's}, a_{\text{Sanborn's}}, i)]$ talks about costs due to passive inventories, and $[BL(\text{Sanborn's}, a, k)]$ figures out balance scores on service and support. It is worthwhile noticing that this last item is currently not extracted as a BAM figure, since it changes very slowly in time with respect to the other variables. The result is presented in equation (5).

$$r(\text{Sell}, a_{\{ \text{Sanborn's} \}_{i}}) = \bigcup_i \left(\{PS(\text{Sanborn's}, i) - CM(a_{\{ \text{Sanborn's} \}_{i}})\} - \{EX(\text{Sanborn's}) + CS(\text{Sanborn's}, \text{Inv}_{\text{passive}})\} + \{BL(\text{Sanborn's}, \text{svc}, \text{spp})\} \right) \\ i \in \dim(\text{BAM}[\text{Sanborn's}, a, k]) \quad (5)$$

A PMA is likely to witness a long series of calculations after invoking a simulation based on these numbers. An example follows².

$$a1: \begin{bmatrix} a2 & 19 & 50 & 8 \\ a2 & 19 & 25 & 8 \\ a2 & 19 & 20 & 8 \\ a3 & 20 & 15 & 8 \\ \dots & \dots & \dots & \dots \\ a5 & 16 & 45 & 3 \\ a5 & 22 & 25 & 12 \\ a5 & 22 & 24 & 16 \\ a6 & 21 & 33 & 8 \\ a6 & 15 & 12 & 33 \\ a7 & 15 & 17 & 28 \\ a7 & 18 & 18 & 34 \\ a8 & 16 & 26 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} -23 \\ 2 \\ 7 \\ 13 \\ \dots \\ -26 \\ 9 \\ 14 \\ -4 \\ 36 \\ 26 \\ 34 \\ -8 \end{bmatrix} \quad (6)$$

² The example shown in (6) is small due to space limitations. The actual database entries and action matrix calculations are large and consume considerable amounts of CPU cycles. This is one of the drawbacks of the method.

In contrast to the computations in (4), in (6) there exists a set of "best" rewards, namely actions by agents $a6$ and $a7$. Therefore, BAM will provide an insight for decision making, given that the system has learned about past events, several steps back. This corresponds to a time trace of the Q-Learning process, as presented in [10]. We now use the time trace to reason backwards and provide a next step into the future. The time trace is shown (in its modified state) in Figure 4.

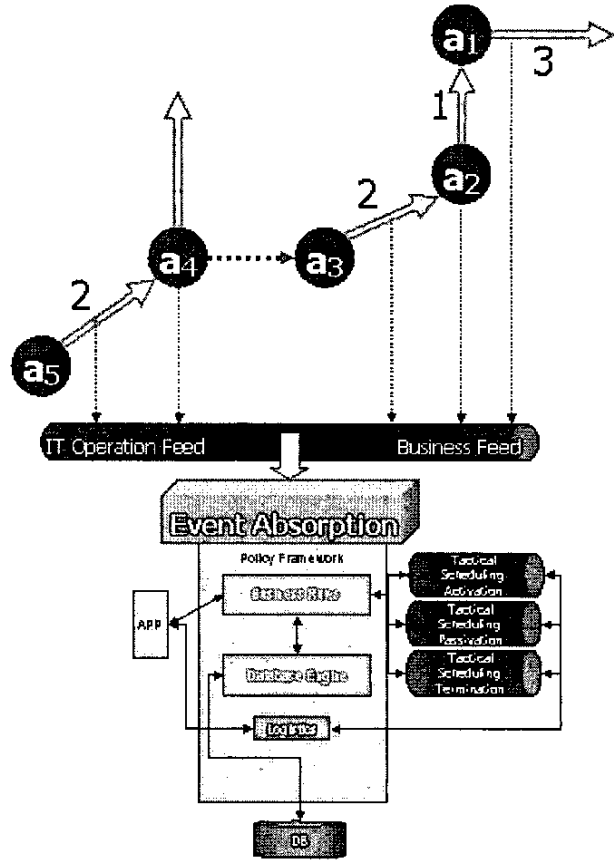


Figure 4, Backwards Reasoning Using Time Trace

Of course, the process is not completely automated. The fine details on what exactly did agents $a6$ and $a7$ did to come out with the "best" rewards may be of qualitative nature, and may correspond to the manager to pinpoint and document. However, the BAM database is connected to other transactional ones, which contain useful data on the orders, payments, inventory management, and other ORMS. Thus, the system is able to show some details on its own. This is shown on the next section.

5 BAM-DSS System for a Sample SMB

We have implemented the above mentioned methodology and framework onto an administrative owned by a SMB manufacturer of cardboard boxes used by pharmaceutical companies to wrap and distribute fragile medicines. We shall name it "EPIX" for confidentiality purposes.

EPIX is a B2B portal. It features standard session management, orders and payment administration, plus supplier catalogs and inventory tracking. EPIX is shown in Figure 5.

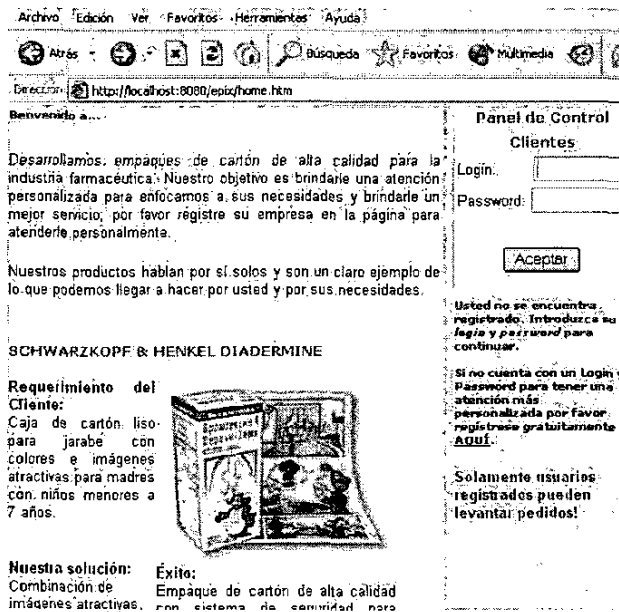


Figure 5, EPIX Main Portal

Our case study consisted on adding BAM and DSS capabilities to the standard web device. The method consisted on adding BAM listeners to a set of hosts, each of which runs a particular administrative system. All host talk to a central server using UDP sockets. BAM commands are interpreted over the UDP stream, and are coded using an XML dialect. The overall system is shown in Figure 6.

BAM listeners are active or down depending on local system administration. The remote communications depend on the central server. The Java programs that implement both sides are low-end code projects that require little deployment and maintenance effort. These programs read local databases and processes, using selected data for building XML streamlets (i.e. jobs) that are sent to the server as soon as it is possible. Following is a segment of the XML code transmitted via UDP.

```
<event date="12.04.2004" site="sales" usr="joe"
client="4405" time="12:45:12" location="perisur_A2">
<event_action name="order placed"><product
sku="33094844" price="1000.00"><product
sku="89394776" price="1100.20"><product
sku="67273989" price="1145.00"></event_action></event>
```

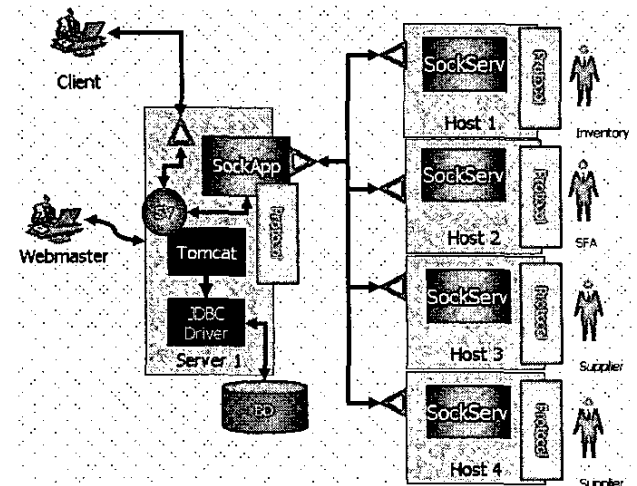


Figure 6, BAM Java System Architecture

The EPIX server calls its remote hosts "status machines". It keeps track of the status of all jobs, and of the hosts. The operator console is shown in Figure 7.

Status Machines			
		Jobs in Queue	Available Capacity
Machine 1	Not Available	100	0
Machine 2	Available	5	95
Machine 3	Available	5	95
Machine 4	Available	5	95
Machine 5	Available	5	95

Figure 7, Operator Console Status Machines

The BAM events are received on the main server and can be visualized, or monitored, using a sniffer facility, shown in Figure 8. Once in the server, internal business logic applies equation (5) calculating the reward for each case. The PMA is not likely to have an interest in viewing raw events. However, the intermediate factors (PS, CM, EX, CS, and BL) plus the final reward comparison, will be useful to her.

Further down the road, DSS must take place at the PMA station. Administrative services are of two types (1) Tactical

and (2) Strategic. Due to space limitations, we will present only one of these in this paper, the Strategic one, because it is closely related to the data produced by the Q-Learning (and Soccer match) metaphor.

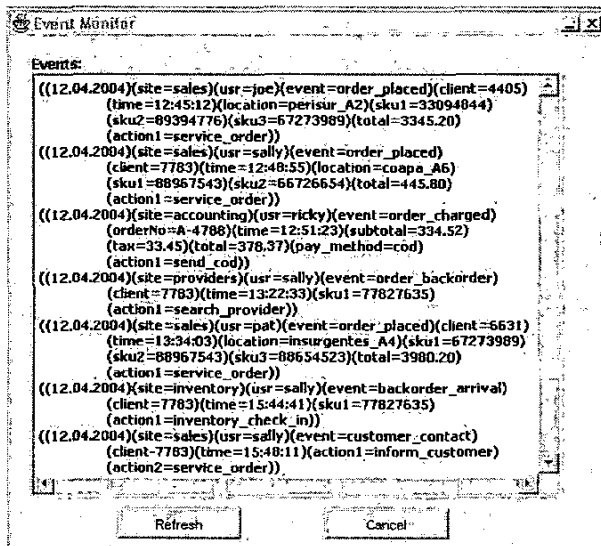


Figure 8, BAM Sniffer Program

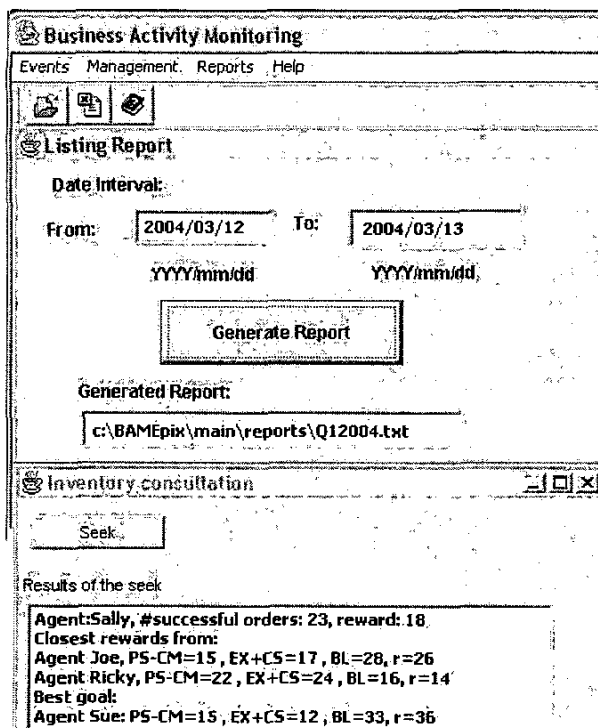


Figure 9, DSS using BAM and Q-Learning

On this Strategic service the PMA is allowed to select one or more administrative areas to report events on. Then she selects a time interval. The system filters and applies business rules shown in Figure 4, coming back with a report like the one shown in Figure 9.

6 Conclusions and further work

We have started the tests on a low-end but cost-effective solution to Business Activity Monitoring on SMB's. The initial results of our tests show that reports like the one in Figure 9 are understood by customers over SMS and PDA email on time-critical missions. This has helped us sell the concept to underteched companies, since little infrastructure is actually needed to deploy these solutions. At the same time, we have started proving that the theoretical grounds laid in [10] actually serve a practical purpose. Finally, we may contribute to making the world of B2B and negotiations a more human one.

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